Chapter 11

System Implementation 1

(Installation & File Conversion)

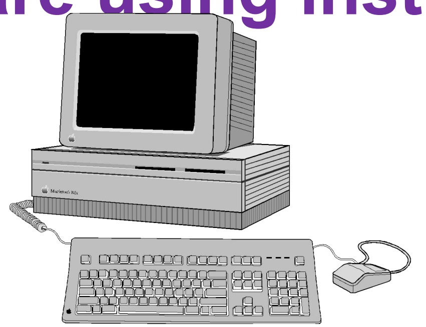
1. Equipment Conversion (Installation)

Installation

- **Areas : Hardware and Software**
- **Amount :**
  - The amount of installation varies from case to case.
  - From almost small (PC system) to installing a big server (mainframe system) and all its peripheral equipment.
- **Examples :**
  - PC system : delivery schedules, networking, application and system software (eg drivers, OS) installation etc
  - Mainframe system : site preparation, hardware

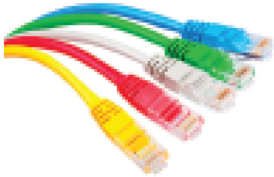
Installation of PCs and Peripherals

- **Small- or medium-size system on established equipment**
  - Negotiating *scheduled run time* and *disk space*.
- **PC systems**
  - **Site planning** : in terms of the availability of *space*, *accessibility*, and *cleanliness*.
  - **Installation** : Install PCs in a small network, and hardware using instruction manuals



Installation of PCs and Peripherals (cont..)

- **Hardware Installation**
  - **Site.** Any office environment with air conditioning.
  - **Power Supply.** Power sockets
  - **Furniture.** Desks and chairs.
  - **Communication.** Telephone to contact for support and advice when face with difficulties.
  - **Cabling.** For network connections
  - **Noise.** Printer room for noisy printers



Installation of PCs and Peripherals

- **Software Installation**
  - Installation of Software. Setup software in hardware. Can be simple by following instructions in the user's manual in minutes.
  - Pre-loaded Software. Most PCs come with pre-loaded OS (Microsoft Windows) and others (eg. Microsoft Office, anti-virus)
  - Software Registration. Online registration is possible to allow online updates.
  - Software Back-ups. Online downloads of software now means backups may not be needed anymore.
  - Software License. Most software requires licenses (one time or annual). Based on the number of users, multi-liceneses can be purchased.

Installation of a Mainframe or Minicomputer

- Many of the issues described above, such as furniture needs, cabling and so on, still apply.
- The particular problems of planning large installation include the following :

1. Site Selection

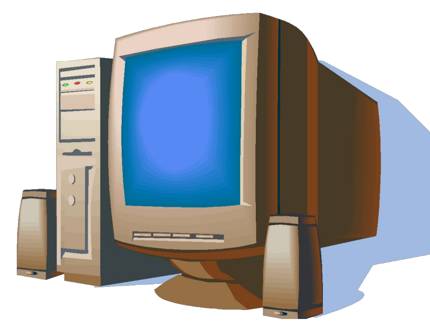
- Site selected for the main computer might be in an existing or a new building.
- Site selection criteria :
  - Space – Adequate space for computer and peripherals, library, stationery, including servicing room.
  - Nearness – to principal user departments.
  - Access – easy access for computer equipment and supplies
  - Expansion – enough room for future expansion.
  - Safety – from fire, nature disasters, attacks etc

2. Site Preparation

- The site preparation may involve consideration of certain potential problems :
  - Furnishings – tables, cabinets etc.
  - Electricity - supplies (eg. 3-phase)
  - Air conditioning - (temperature, humidity and dust).
  - Raised floor - (or false ceiling) so that cables may pass from one piece of equipment to another and to prevent generation of static electricity.
  - Fire protection - devices eg. sprinkle

3. Standby Equipment

- Standby equipment should be arranged, to ensure continuity of processing in the event of power of computer failure, breakdowns etc.
- Examples : Such equipment may include standby generators and standby computers (servers).



Exercise for Students

- XYZ Ltd has just completed its UAT and are now ready for Equipment Conversion. Discuss TWO (2) things which need to be carried out before Equipment Conversion and TWO (2) things which need to be carried out during Equipment Conversion

Suggested Solution

- TWO (2) things before Equipment Conversion :
  - Site selection : site for server and clients
  - Site preparation : furnishing, electricity, air-conditioning, fire-protection
- TWO (2) things during Equipment Conversion :
  - Hardware Installation : Servers, Clients, Devices, Network Infrastructure
  - Software installation : OS, DBMS, Application, drivers

2. File Conversion and Creation

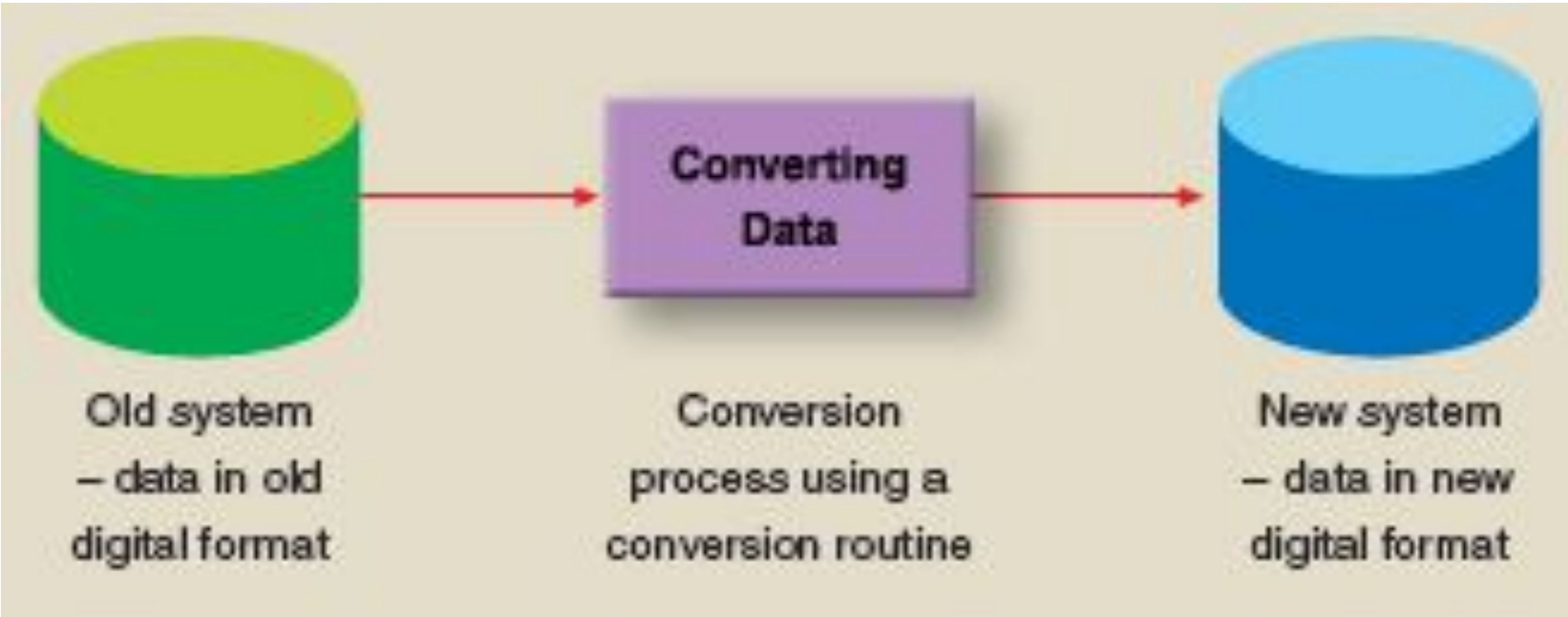
Objectives of File Conversion and Creation

- **Accuracy** - produce a file free from errors.
- **Completeness** - ensure that the process is complete and a success, otherwise subsequent computer file updating will be greatly hampered.
- **Medium** - produce a file on a suitable specific storage medium eg disk.

Data Conversion

- **Meaning.** Data from the old system is converted to new system. This can be carried out using conversion routine.
- **Data mapping.** Data files between the old and the new systems will be mapped (data fields).
- **Conversion routines.** When implementing new software application data has to be converted from the old system using data conversion routines (a program).

Data Conversion

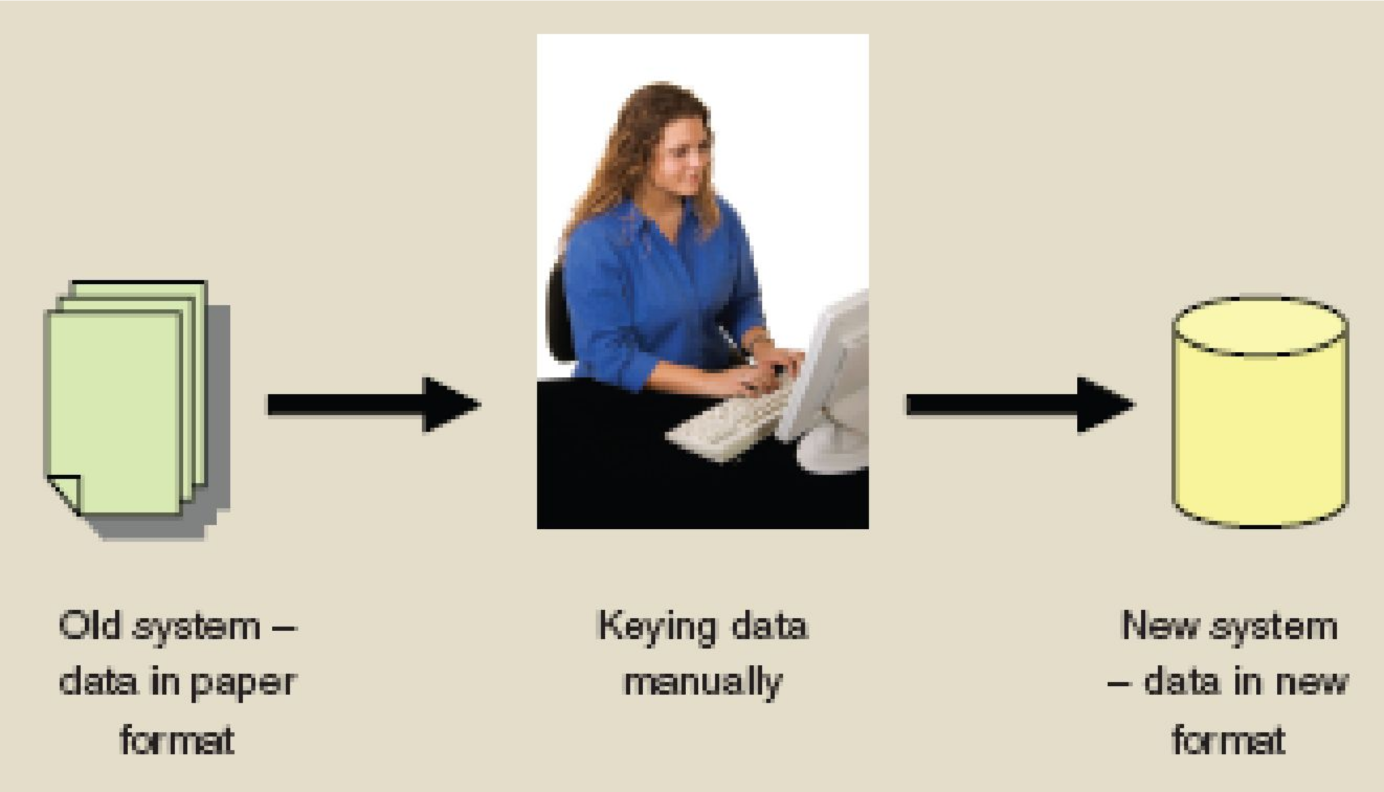


Both the old and new systems with different data formats

Data Creation

- Meaning. New data not found in old system will be created in the new system using creation routine for manual data entry.
- Creation routines. This program is developed for manual data entry purpose.
- Data validation. Data will have to be validated to help ensure that incorrect data is not entered into the new system eg. character check, size check.

Data Creation

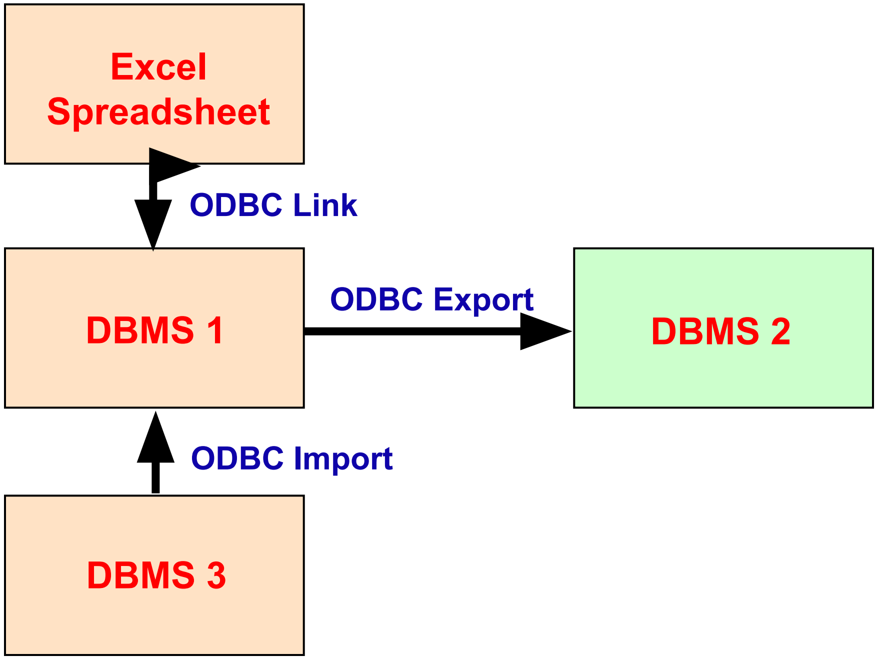


Data entry – a manual process of entering data from a paper-based system

Methods of Converting Data

- **Automated approach** - where data are of different format between old and new systems, the data must be convert in standard format (ASCII, ODBC) first.
- **Manual keying** - a difficult process of converting data into a new system from a manual system.

ODBC (Open Database Connectivity) – an industry-standard protocol that allows DBMSs from various vendors to interact and exchange data.

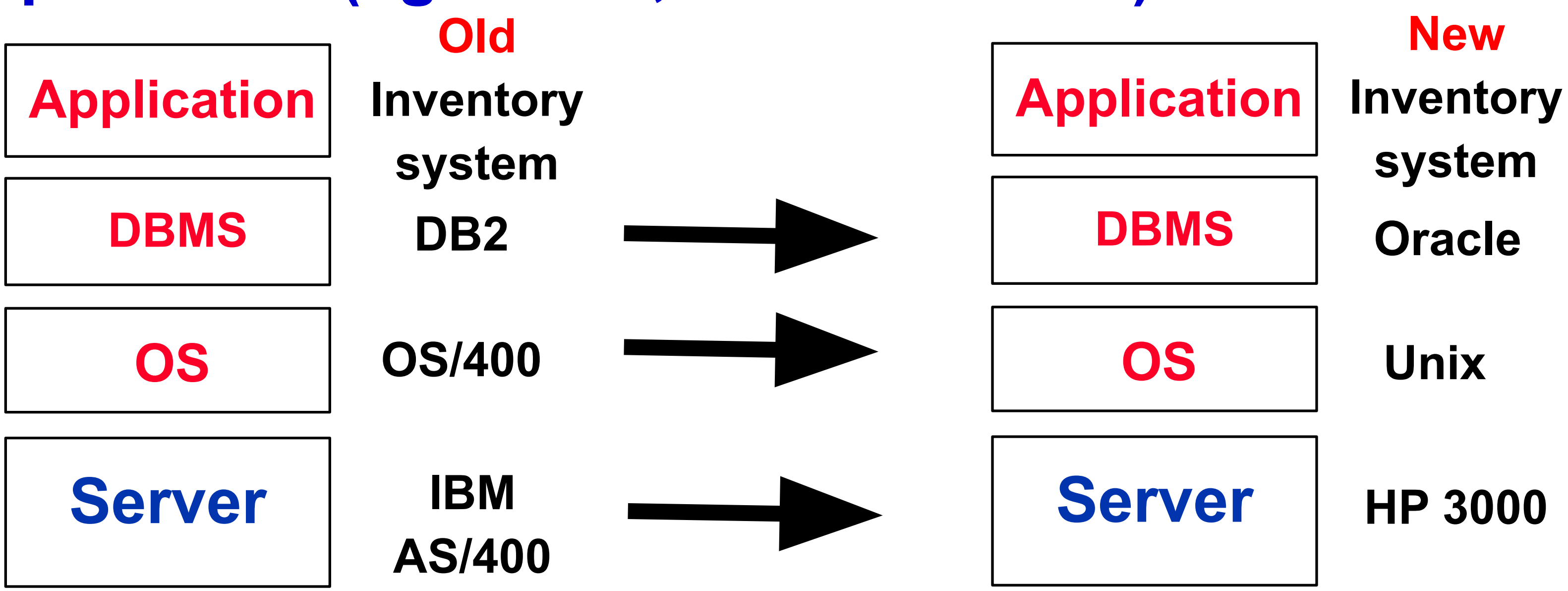


Common Data Conversion Problems

(for computerized systems)

1. Technical incompatibility of the two systems

- **Meaning** - when new software runs on a different hardware and software. It will not be technically feasible to covert the data. One solution is to print out the data and entered manually into the new system.
- **Example** - old and new systems on 2 different platforms (eg server, OS and DBMS).



2. Problems of new data fields

- **Meaning** - data fields in the proposed system but are not in the current system due to emerging new requirements.
- **Example** - the proposed CUSTOMER data file includes *gender* or *Date of Birth*, a field not held on the current computer system.

Current System	Proposed System
Customer ID :	Customer ID :
Name :	Name :
Tel No. :	Tel No. :
Email :	Email :
	Date of Birth:

3. Problems of matching data fields

- **Meaning** - no exact match between the data fields on the current and proposed systems. Fields may need to be combined or split up.
- **Examples** – 'delegate-name' and 'address'.

Current System	Proposed System
Address:	House No. :
	Street Name
	State :
	Postcode:

4. Problems of new data files

- **Meaning** - **new system will have extra files. Appropriate values of the fields in these files have to be determined.**
- **Example** - **entry of examiner codes into the new EXAMINER file and populate this field in the EXAMINATION file.**

Current System	Proposed System
Registration →	Registration
Reservation →	Reservation
Borrowing →	Borrowing
Returning →	Returning
	Cancellation (NEW)

5. Problems of different field lengths

- **Meaning** - transfer of data with different field lengths
- **Examples** - Examiner names have been abbreviated and now need to be expanded, telephone numbers (from 7 digits to 8 digits)

Current System	Proposed System
Tel No. : 3237567	Tel No. : 32375674

ASSESSMENT file **Old File**

Field name	Length of field	Type of field
Examination number	4	Numeric
Examination name	30	Character
Examiner name	30	Character
Examiner address	50	Character
Passmark	2	Numeric

EXAMINATION file **New File 1**

Field name	Length of field	Type of field
Examination number	4	Numeric
Examination name	30	Character
Examiner code	3	Numeric
Syllabus summary	100	Character
Passmark	2	Numeric

EXAMINER file **New File 2**

Field name	Length of field	Type of field
Examiner code	3	Numeric
Examiner name	50	Character
Examiner address line 1	20	Character
Examiner address line 2	20	Character
Examiner address line 3	20	Character

Exercise for Students

Identify 3 issues or problems when populating the new file structure ?

Suggested Answers

1. How to split up the Examiner Address to three address lines. (Unmatched Data Fields)
2. How to organise the entry of Examiner Codes and data into the new EXAMINER file and new EXAMINATION file. (New Data Files)
3. Whether Examiner Names have been abbreviated in the current ASSESSMENT file and now need to be expanded. (Different Field Length)
4. How to enter Syllabus Summary information into the new system. This data does not exist on the old system. (New Data Field)

Data Conversion Problems (for manual system)

- 1. Set Up of Master Files
  - The task of file creation can be a daunting task.
  - The resources of the department may not be sufficient or willing to undertake such activities.
- 2. Entry of System Derived Fields
  - File creation programs may have to be written to automatically capture historical information into the system.
  - Example - the date-of-last-order may not be available at time of receipt of order.
- 3. Lack of Historical Data Restrict Use
  - Reports running off the date-of-last-order introduced in the previous section, may be of little use until the second or third year of the system's use.

File Conversion Plan

- Set a timetable for conversion.
- Carry out a dummy run
- Decide upon a cut-off date
- Training of relevant staff
- Run the data validation programs
- Try to plan for the quickest possible changeover

What is a dummy run ?

What is a cut-off date ?